Shane Sarosh

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Education

Vellore Institute of Technology (VIT) — Vellore, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.79/10 — *Aug. 2024 – May 2028*

Experience

The South Indian Bank Limited — Kochi, India
AI Intern — *May 2026 – June 2026*

- Built NoteForge, an AI-assisted QA platform that converts banking business-requirement documents into structured, traceable test cases using FastAPI, Pydantic, and LLM-based generation.
- Validated the system against digital-banking workflows spanning national eKYC identity verification (Aadhaar), core-banking integration (Finacle), and real-time payment rails (UPI/IMPS/NEFT/RTGS), including transaction limits and failure handling.

Research & Projects

Ask r/Vit: Retrieval-Augmented Question Answering | *Python, OpenAI API, SQLite FTS5* — Nov. 2026

- Built a cited question-answering system over the entire r/Vit subreddit (55,919 threads, 485K comments), fusing BM25 keyword and embedding search via reciprocal-rank fusion, followed by LLM re-ranking.
- Extracted 102,568 structured facts with an LLM pass on the OpenAI Batch API, using a resumable, stall-tolerant job pipeline; total build cost about $31.
- Built a 430-question evaluation set and ablated retrieval designs (HyDE, re-rank pool size, whole-thread context): 95% of answers correct or partially correct, with the source thread retrieved for 96% of questions.
- Added an agentic deep-research mode (multi-query search, whole-thread reading, cross-checking) with a hard per-query cost cap; a typical run does 15 searches over 50+ threads for about $0.10.

Searchless Chess: Mechanistic Interpretability | *Python, JAX, Scikit-learn* — GitHub | Jun. 2026

- Probed eight chess concepts across all 17 layers of DeepMind's 270M-parameter Searchless Chess transformer, training 136 linear probes on 5,488 positions.
- Recovered board state with near-perfect fidelity from layer activations (material balance $R^2 = 0.999$, center control $R^2 = 0.989$, near-ceiling castling-rights AUC).
- Found that this interpretable structure peaks in layers 1–3 before degrading as later layers shift to encoding action values.
- Built a Chrome extension and WebSocket pipeline that visualizes the transformer's move preferences in real time during post-game analysis.

Hybrid Job Hunter | *Python, Playwright, GitHub Actions, Telegram API* — GitHub | Jul. 2026

- Engineered adapters for seven ATS and career platforms, combining structured APIs with Playwright browser automation to monitor JavaScript-rendered career pages on scheduled GitHub Actions runs with no dedicated infrastructure.
- Designed for fault tolerance — per-source health tracking, request budgets, and queued retries — running unattended on a scheduled cadence.

Technical Skills

Programming: Python, SQL, C++, JavaScript
Data Analysis: Pandas, NumPy, Matplotlib, exploratory data analysis, data cleaning, data visualization
Machine Learning: scikit-learn, JAX, RAG, embeddings, LLM evaluation, model evaluation, linear probing
Development Tools: Git, FastAPI, Pydantic, SQLite, Playwright, GitHub Actions

Certifications

Microsoft Certified: Azure AI Fundamentals | *Microsoft* — Credential
Qiskit Global Summer School 2026 - Quantum Excellence | *IBM* — *Aug. 2026*